

Drug repurposing strategy: An emerging approach to identify potential therapeutics for treatment of bovine mastitis

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ABSTRACT

The current study was designed to characterize methicillin-resistant *Staphylococcus aureus* (MRSA) isolated from bovine milk, along with its response to antibiotics, and ultimately reverse its mechanism of resistance by modulation with non-antibiotics. The synergistic combination of antibiotics with NSAIDs were tested *in-vivo* by giving MRSA challenge to rabbits. The current study reported an overall 23.79% prevalence of MRSA. The BLAST alignment of current study sequences revealed 99% similarity with *mecA* gene of MRSA from NCBI database. The current study isolates were more similar to each other and also with reference sequences as compared to other *mecA* gene sequences from Turkey, India, and Russia. Antibigram of MRSA isolates showed a highly resistant response to cefoxitin, amoxicillin, and gentamicin. Amoxicillin, gentamicin, tylosin, vancomycin, and ciprofloxacin elicited a significant response ($p < 0.05$) in combination with non-antibiotics against tested MRSA isolates. The highest zone of inhibition (ZOI) increase was noted for vancomycin in combination with flunixin meglumine (145.45%) and meloxicam (139.36%); gentamicin with flunixin meglumine (85.71%) and ciprofloxacin with ivermectin (71.13%). Synergistic behavior was observed in the combination of gentamicin with ketoprofen; sulfamethoxazole and oxytetracycline with meloxicam. Hematological analysis showed significant differences ($p < 0.05$) among lymphocyte count and bilirubin. On histopathological examination of skin tissue, hyperplasia of epithelium, sloughed off epidermis, hyperkeratosis, infiltration of inflammatory cells, and hemorrhages were observed. The highest cure rate was observed in case of gentamicin in combination with ketoprofen as compared to other treatment groups. The current study concluded antibiotics in combination with non-antibiotics as potential therapeutic agents for resistance modulation against MRSA. This study will help to devise treatment and control strategies against bovine mastitis. Although the prospect of using NSAIDs to manage infections caused by MRSA appears to be a promising direction, further studies should be conducted to test these medications using suitable *in-vivo* models in controlled clinical trials to justify their repurposing as a treatment for MRSA infections.

1. Introduction

Staphylococcus aureus (*S. aureus*) is one of the most commonly isolated mastitogens around the globe [1]. *S. aureus* has developed resistance to antibacterial agents through restriction of drug absorption, drug targets, and receptors modification, drug enzyme inactivation, and drug outflow pathways. In particular, the localization of resistance encoding genes on mobile genetic islands such as transposons, plasmids and altering genetic constituents favors the horizontal as well as a vertical transfer of resistance between interacting bacterial species [2].

Methicillin-resistant *S. aureus* (MRSA) is a highly resistant strain isolated from bovine mastitis that usually develops resistance by acquiring *mecA* gene, which encodes bacterial cell wall PBP2 α protein [3].

MRSA strains of both animal and human origin are thought to serve as an important reservoir of antibacterial resistance encoding genes [4, 5]. Livestock-originated multidrug-resistant *S. aureus* strains possess certain genes including *mecA*, *aacA-D*, *tetK* and *tetM*, *ermA* and *ermC*, *msrA*, *linA* and *vatA*, *vatB* and *vatC* responsible for the development of antibiotic resistance against methicillin, aminoglycosides, tetracyclines, macrolide–lincosamide–streptogramin B, macrolides, lincosamides, and

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Table 1
Prevalence of subclinical mastitis and *S. aureus*.

District	Animal	No. of Milk samples	SCM Positive	<i>S. aureus</i> Positive	MRSA	
					Phenotypic (%)	Genotypic (%)
Faisalabad	Cattle	192	141	79	42 (53.16)	23 (29.11)
	Buffalo	192	124	64	25 (39.06)	14 (21.88)
Rawalpindi	Cattle	192	129	58	24 (41.38)	14 (24.14)
	Buffalo	192	114	47	19 (40.43)	08 (17.02)
Total		768	508	248 (48.82)	110 (44.35)	59 (23.79)

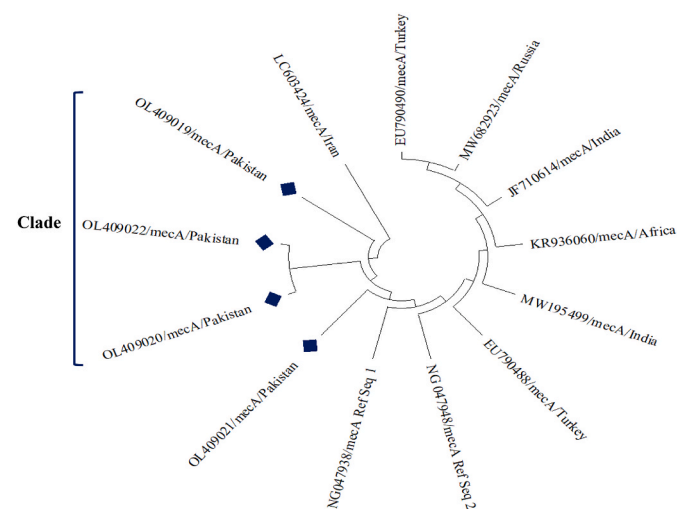


Fig. 1. Phylogenetic tree constructed by maximum likelihood method indicating the *mecA* gene similarity among current and reported genes.

Table 2
Antibiogram of MRSA isolates.

Antibiotics	MRSA (n = 13)		
	Sensitive (%)	Intermediate (%)	Resistant (%)
Cefoxitin (30 µg)	07.69	0	92.31
Amoxicillin (10 µg)	15.38	0	84.62
Gentamicin (10 µg)	07.69	0	92.31
Amikacin (30 µg)	84.62	15.38	0
Ciprofloxacin (5 µg)	53.85	30.77	15.38
Levofloxacin (5 µg)	76.92	07.69	15.38
Moxifloxacin (5 µg)	84.62	15.38	0
Vancomycin (30 µg)	07.69	30.77	61.54
Oxytetracycline (30 µg)	38.46	0	61.54
Tylosin (30 µg)	15.38	23.08	61.54
Chloramphenicol (10 µg)	61.54	15.38	23.08
Trimethoprim + Sulfamethoxazole (1.25 µg + 23.75 µg)	23.08	07.69	69.23
Fusidic acid (10 µg)	53.85	07.69	38.46
Linezolid (30 µg)	100	0	0

streptogramin A groups of antibiotics, correspondingly [6]. This resistance characteristic not only increases pathogen combating costs but also exploits the risk of such superbugs being transmitted to human beings through the consumption of animal products or handling of contaminated materials [7].

In mastitis therapeutics, there is increasing awareness of the use of non-antibiotics including non-steroidal anti-inflammatory drugs (NSAIDs) in cattle to manage the inflammation, pain, and endotoxin-induced effects such as fever that accompany opportunistic bacterial infections [8,9]. Although the antimicrobial mechanism of action of various non-antibiotic agents is not clear, it can be supposed that these perform such functions by modifying cell permeability, efflux pump of

Table 3
Antibacterial activity of tested drugs against MRSA isolates.

Group	Drug's Name	Zone of inhibition (mm)	p-value
Antibiotics	Amoxicillin	15.00 ± 1.00	0.017
	Gentamicin	14.00 ± 2.645	
	Sulfamethoxazole	11.00 ± 1.04	
	Tylosin	15.67 ± 1.527	
	Oxytetracycline	14.00 ± 2.645	
	Chloramphenicol	15.53 ± 3.722	
	Vancomycin	11.00 ± 1.480	
	Ciprofloxacin	15.00 ± 2.80	
Non-antibiotics	Meloxicam	9.13 ± 1.026	0.001
	Diclofenac sodium	11.53 ± 1.747	
	Flunixin meglumine	13.33 ± 1.527	
	Ketoprofen	10.40 ± 0.964	
	Ivermectin	5.67 ± 1.527	

$p < 0.05$ show significant variation.

bacteria, ion transporter, and by disturbing the action of vital enzymes [10]. NSAIDs, including carprofen, work by interfering with bacterial DNA replication and repair by snooping with DNA polymerase, because DNA polymerase may be an evolving antimicrobial site involved in the replication of DNA [11]. Diclofenac shows antibacterial properties by disrupting bacterial DNA ultimately by using deoxythymidine uptake [12], while selective COX-2 inhibitors exhibited antimicrobial action through an upsurge of lipoxygenase pathway and promoting leukotriene B₄, leukotriene D₄, and cytokine expression as well as improved bacterial clearance [13].

Histopathological analysis is an important tool for evaluating tissue damage from bacterial invasion. However, the drug candidates uncovered by *in-vitro* screening techniques may have different therapeutic effects when subjected to *in-vivo* assay which is due to different drug membrane permeability, metabolism, and the participation of the host immune system [14]. Therefore, the current study was planned to investigate the antimicrobial synergy of antibiotic and non-antibiotic drugs against molecularly characterized MRSA isolates from bovines through *in-vitro* and *in-vivo* assays. Moreover, the hematobiochemical and histopathological alterations in MRSA challenged rabbit model was also assessed.

2. Materials and methods

2.1. Isolation and identification of MRSA from bovine milk

A total of 508 subclinical mastitis-positive milk samples from bovine origin were processed for isolation of *S. aureus* microbiologically and biochemically [15]. Phenotypic confirmation of *S. aureus* was done by colony characteristics, Gram's staining, and various biochemical tests [16] while genotypic confirmation was done through PCR by targeting the *nuc* gene [17].

Phenotypic MRSA identification was done by Oxacillin diffusion assay as described by the Clinical and Laboratory Standard Institute (CLSI) [18,19]. The DNA was extracted from the phenotypically confirmed MRSA through a bacterial DNA extraction kit (WizPrep™ gDNA Mini Kit). PCR was carried out for molecular confirmation of

Table 4

Comparison of zones of inhibitions (mm) of MRSA isolates against different combinations of antibiotics/drugs.

Drugs/Antibiotics used		Mean $\pm$ Std.	p-value
Drug's Name	Combination of Drugs		
Amoxicillin	Alone	15.00 $\pm$ 1.00	0.000
	Amoxi + Ivermectin	23.67 $\pm$ 1.527	
	Amoxi + Meloxicam	19.67 $\pm$ 1.527	
	Amoxi + Ketoprofin	21.33 $\pm$ 0.577	
	Amoxi + Flunixin	21.00 $\pm$ 1.00	
	Amoxi + Diclofenac	23	
Gentamicin	Alone	14.00 $\pm$ 2.645	0.000
	Genta + Ivermectin	21.00 $\pm$ 1.00	
	Genta + Meloxicam	25.33 $\pm$ 0.577	
	Genta + Ketoprofin	22.67 $\pm$ 2.08	
	Genta + Flunixin	26.00 $\pm$ 1.4	
	Genta + Diclofenac	22.33 $\pm$ 1.527	
Sulfamethoxazole	Alone	11.00 $\pm$ 1.04	0.082
	Sulfa + Ivermectin	11.00 $\pm$ 3.00	
	Sulfa + Meloxicam	18.67 $\pm$ 3.785	
	Sulfa + Ketoprofin	16.33 $\pm$ 5.033	
	Sulfa + Flunixin	16.33 $\pm$ 3.055	
	Sulfa + Diclofenac	18.0 $\pm$ 4.582	
Tylosin	Alone	15.67 $\pm$ 1.527	0.000
	Tylo + Ivermectin	23.00 $\pm$ 1.44	
	Tylo + Meloxicam	24.00 $\pm$ 1.60	
	Tylo + Ketoprofin	25.33 $\pm$ 0.577	
	Tylo + Flunixin	21.67 $\pm$ 2.082	
	Tylo + Diclofenac	21.33 $\pm$ 1.527	
Oxytetracycline	Alone	14.00 $\pm$ 2.645	0.079
	Oxy + Ivermectin	10.67 $\pm$ 1.527	
	Oxy + Meloxicam	15.00 $\pm$ 3.020	
	Oxy + Ketoprofin	11.13 $\pm$ 1.665	
	Oxy + Flunixin	12.50 $\pm$ 1.50	
	Oxy + Diclofenac	10.33 $\pm$ 1.222	
Chloramphenicol	Alone	15.53 $\pm$ 3.722	0.504
	Chlor + Ivermectin	10.87 $\pm$ 3.384	
	Chlor + Meloxicam	16.27 $\pm$ 4.026	
	Chlor + Ketoprofin	17.46 $\pm$ 4.105	
	Chlor + Flunixin	15.00 $\pm$ 2.291	
	Chlor + Diclofenac	17.00 $\pm$ 2.887	
Vancomycin	Alone	11.00 $\pm$ 1.480	0.003
	Vanco + Ivermectin	23.00 $\pm$ 1.00	
	Vanco + Meloxicam	26.33 $\pm$ 1.527	
	Vanco + Ketoprofin	23.24 $\pm$ 2.645	
	Vanco + Flunixin	27.00 $\pm$ 2.646	
	Vanco + Diclofenac	23.00 $\pm$ 2.00	
Ciprofloxacin	Alone	15.00 $\pm$ 2.80	0.001
	Cipro + Ivermectin	25.67 $\pm$ 1.527	
	Cipro + Meloxicam	24.98 $\pm$ 1.00	
	Cipro + Ketoprofin	24.00 $\pm$ 2.646	
	Cipro + Flunixin	22.00 $\pm$ 1.064	
	Cipro + Diclofenac	25.67 $\pm$ 2.082	

p < 0.05 show significantly different effect.

MRSA by targeting the *mecA* gene as P1: TGGCATTCGTGTGACAATCG as forward and reverse primer P2: CTGGAACCTGTTGAGCAGAG at 310bp [20]. PCR positive samples were sent for sequencing to 1st Base Molecular Laboratory, Singapore. These sequences were submitted to the NCBI database and got the following accession numbers (OL409019, OL409020, OL409021, and OL409022) which were further used in phylogenetic analysis.

2.2. Molecular characterization of MRSA

MRSA isolates were characterized for *mecA* gene by performing phylogenetic analysis using suitable bioinformatics tools [21]. The sequenced isolates of MRSA were subjected to a Basic local alignment search tool (BLAST) to check the similarity of the study isolates with already reported *mecA* gene sequences. The *in-silico* validation of study

gene sequences for *mecA* was conferred by its resemblance with NCBI reference sequences of *mecA* gene at alignment. The software (Bioedit Sequence Alignment Editor Version 7.2.5) was used for *Clustal W* multiple alignment of study and reported gene sequences. The evolutionary relatedness of *mecA* gene sequences was further explored by constructing the phylogenetic tree at the phylogeny test of 1000 bootstrap replications by using maximum likelihood method through Mega X software.

2.3. In-vitro drug therapeutics of MRSA isolates against antibiotics and non-antibiotics

The *in-vitro* susceptibility of characterized MRSA against selected antibiotics and non-antibiotics was evaluated using disc diffusion, well diffusion, and broth microdilution assays.

2.3.1. Antibigram of MRSA

The selected antibiotics including Levofloxacin (5 μ g), Linezolid (30 μ g), Tylosin (30 μ g), Gentamicin (10 μ g), Amikacin (30 μ g), Ciprofloxacin (5 μ g), Moxifloxacin (5 μ g), Trimethoprim + Sulfamethoxazole (25 μ g), Oxytetracycline (30 μ g), Ceftiofur (30 μ g), and Fusidic acid (10 μ g) were used for this trial [19]. Antibiotic discs were applied aseptically and the zones of microbial growth inhibition were compared with standard zones to differentiate MRSA strains as sensitive, intermediate, and resistant [18].

2.3.2. Well diffusion assay of antibiotics and non-antibiotics against MRSA

Antibiotics and non-antibiotics serial dilutions, alone and in combination, were poured into the wells, made in pre-seeded Mueller Hinton agar plates, and incubated overnight at 37 °C. Microbial growth inhibition zones were measured [22]. The enhanced zone of inhibition (ZOI) in combination as compared to the individual drug was considered effective.

2.3.3. Minimal inhibitory concentration determination against MRSA

Minimal inhibitory concentrations (MIC) of antibiotics and non-antibiotics against tested microorganisms were determined by broth micro-dilution using a 96 wells titration plate. Serial dilutions of tested agents were done in the wells containing nutrient broth except for control wells. OD value at 570 nm was taken before and after overnight incubation at 37 °C [23].

2.3.4. Synergy testing of drugs against MRSA

Synergy testing of antibiotic and non-antibiotic combinations against MRSA was done by checkerboard method. Serial dilutions of tested agents were done in the wells containing nutrient broth except for control wells. OD value at 570 nm was taken before and after overnight incubation at 37 °C [24]. Fractional inhibitory concentration indices were calculated by the formula given below [22].

$$FICI = FIC \text{ of product A} + FIC \text{ of product B}$$

$$FIC \text{ of product A} = \frac{\text{MIC of product A in combination with product B}}{\text{MIC of product A alone}}$$

$$FIC \text{ of product B} = \frac{\text{MIC of product B in combination with product A}}{\text{MIC of product B alone}}$$

2.4. In-vivo evaluation of combinations of antibiotics with non-antibiotics in rabbits

Successful combinations of antibiotics and non-antibiotics from *in-vitro* experiments were evaluated in the *in-vivo* experiment. Rabbits of almost equal bodyweight and age were purchased from a reliable breeding center. An aliquot of 50 μ l MRSA (10^8 CFU/mL in phosphate-buffered saline) suspension was inoculated into the skin of rabbits. An antibiotic combination dose was applied on the appearance of signs and was given every 12 h for 5 days. The rabbits were kept for 1 week after

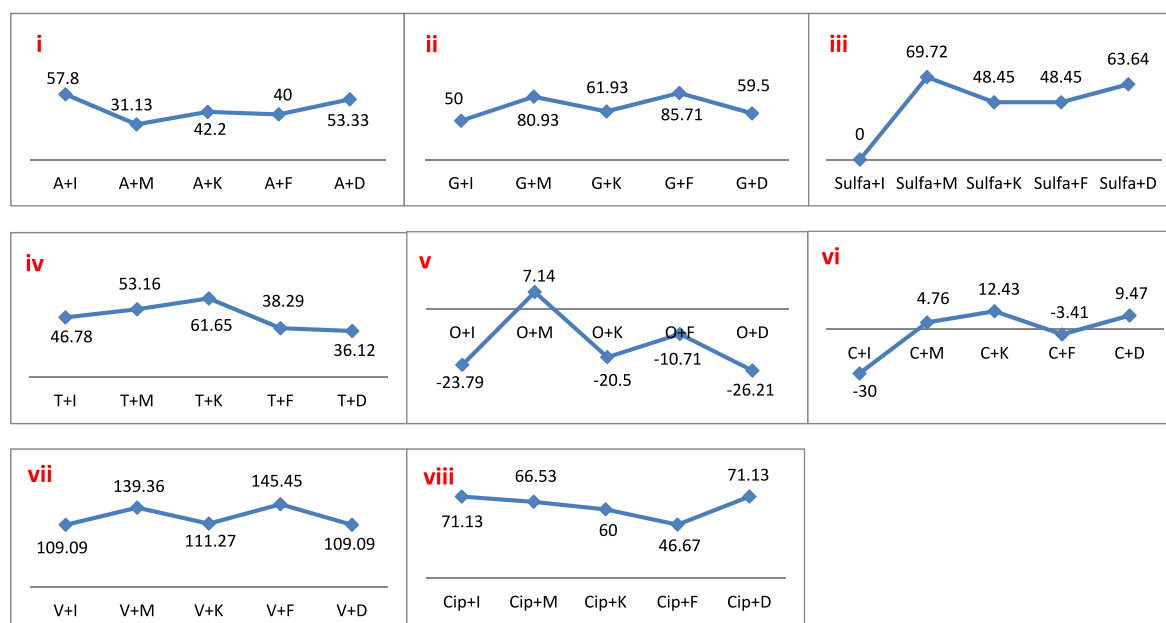


Fig. 2. Variation (increase/decrease) in zone of inhibition (ZOI) of antibiotics when compared with their combinations with non-antibiotics (i)—% variation in ZOI of Amoxicillin (A) when it was compared with its combination with non-antibiotics; (ii)—% variation in ZOI of Gentamicin (G) when it was compared with its combination with non-antibiotics; (iii)—% variation in ZOI of Sulfamethoxazole (Sulfa) when it was compared with its combination with non-antibiotics; (iv)—% variation in ZOI of Tylosin (T) when it was compared with its combination with non-antibiotics; (v)—% variation in ZOI of Oxytetracycline (O) when it was compared with its combination with non-antibiotics; (vi)—% variation in ZOI of Chloramphenicol (C) when it was compared with its combination with non-antibiotics; (vii)—% variation in ZOI of Vancomycin (V) when it was compared with its combination with non-antibiotics; (viii)—% variation in ZOI of Ciprofloxacin (Cip) when it was compared with its combination with non-antibiotics.

the last injection [25]. The interpretation of drug efficacy was aided by wound healing patterns along with hematobiochemical changes in rabbits on subsequent days. White blood cells (WBCs), red blood cells (RBCs), hemoglobin, and lymphocytic values at days (0, 3, 6, 9, 12, and 15th) were analyzed by using a hematological analyzer [26]. The comparative histopathology of skin tissue was also observed. H and E staining of these tissue sections were performed following the method suggested by Ref. [27].

2.5. Statistical analysis

The prevalence of *S. aureus* was calculated as per the formula described by Ref. [28]. The susceptibility trials were analyzed by descriptive statistics while combination drug trials were evaluated by ANOVA technique. Percentage increase in the zone and fractional inhibitory indices were calculated using the formulas given by Ref. [22]. The significance was checked at 5% probability ($p < 0.05$) using SPSS (version 20) of the computer program.

3. Results

3.1. Prevalence of *S. aureus* and MRSA

A total of 508 subclinical mastitis-positive milk samples from bovine origin were processed for isolation and identification of *S. aureus*, out of which 248 (48.82%) isolates were obtained. The current study exhibited an overall prevalence of phenotypic MRSA 44.35% and a genotypic prevalence of MRSA 23.79% in the study area (Table 1).

3.2. MRSA characterization by *mecA* gene

The BLAST alignment of current study sequences revealed 99% similarity with *mecA* gene of MRSA from NCBI database. The comparison of retrieved NCBI reference sequences of *mecA* gene with current study sequences showed a high resemblance at the nucleotide level. The

Clustal W multiple alignment with reference and reported sequences indicated the high similarity except at some nucleotide positions (Fig. S1). However, the study isolates also exhibited differences among themselves at nucleotide comparison. The isolate HB10 (OL409020) showed substitution with cytosine base both at position 2 and 3 along with the addition of adenine at position 275 with respect to sequence HB4 (OL409019). Moreover, the substitution of cytosine and guanine at positions 1st, 2nd, and of adenine both at position 4th and 6th were noted in isolate HC10 (OL409021). Furthermore, the deletion at position 11th, 14th, 40th, 41st, 42nd, 62nd, 103rd, 104th, 236th, 268th and 276th were commonly seen in all study isolates. The comparison with reference and already reported sequences showed common substitution with adenine at position (2nd, 3rd, 8th, 11th, 14th, 39th and 275th), Thymine (4th), Guanine (6th), and Cytosine (7th) positions respectively (Fig. S1). However, similar deletion at position 40th 41st, 42nd, 183rd, 184th, 236th, 268th and 276th was observed in all *mecA* gene sequences. While the gene sequence of (Accession number: LC603424) from Iran showed the least similarity and exhibited difference at more than a hundred positions respectively (Fig. S1).

The phylogenetic tree also exhibited more evolutionary relationships of study isolates in comparison with other isolates (Fig. 1). However, the isolate HC10 (OL409021) showed a somewhat more evolutionary resemblance with NCBI reference sequences (1 and 2). The current study isolates were more similar to each other and also with reference sequences as compared to other *mecA* gene sequences from Turkey, India, and Russia. While the sequence from Iran forms the out-group and exhibits the least similarity with current and reported sequences of *mecA* gene from other countries (Fig. 1).

3.3. Antibigram of MRSA isolates

Disc diffusion assay showed that isolates elicit highly resistant response to cefoxitin (92.31%), amoxicillin (84.62%), and gentamicin (92.31%) while showed highest susceptibility towards linezolid (100%), moxifloxacin (84.62%), and levofloxacin (76.92%) (Table 2). Overall

Table 5

Synergy testing of drugs against MRSA isolates.

Combinations	MIC A	MIC AB	FIC A	MIC B	MIC BA	FIC B	FICI	Results
Amoxi + Meloxi	15.625	8.859	0.567	84.822	30.907	0.364	0.931	Additive
Amoxi + Ketoprofen	12.109	6.844	0.565	70.871	54.663	0.771	1.336	Indifferent
Amoxi + Flunixin	15.625	7.813	0.5	125	59.927	0.479	0.979	Additive
Amoxi + Diclofenac	15.625	14.648	0.938	77.247	117.188	1.517	2.455	Indifferent
Amoxi + Ivermectin	12.109	15.625	1.290	37.464	54.663	1.459	2.749	Indifferent
Genta + Meloxi	7.813	3.906	0.5	117.188	37.464	0.319	0.819	Additive
Genta + Ketoprofen	15.625	3.906	0.25	145.833	35.435	0.243	0.493	Synergistic*
Genta + Flunixin	3.906	5.857	1.499	70.871	62.5	0.882	2.381	Indifferent
Genta + Diclofenac	6.569	3.906	0.595	77.247	59.927	0.776	1.371	Indifferent
Genta + Ivermectin	3.906	1.953	0.5	34.169	77.247	2.261	2.761	Indifferent
Sulfa + Meloxi	31.25	6.844	0.219	145.833	37.464	0.257	0.476	Synergistic*
Sulfa + Ketoprofen	15.625	13.055	0.836	77.247	125	1.618	2.454	Indifferent
Sulfa + Flunixin	12.109	14.648	1.209	70.871	54.663	0.771	1.980	Indifferent
Sulfa + Diclofenac	12.109	6.569	0.543	125	40.994	0.328	0.871	Additive
Sulfa + Ivermectin	15.625	17.932	1.148	54.663	72.917	1.333	2.481	Indifferent
Tylo + Meloxi	12.109	13.055	1.078	55.768	69.671	1.249	2.327	Indifferent
Tylo + Ketoprofen	15.625	17.932	1.148	41.016	62.5	1.524	2.672	Indifferent
Tylo + Flunixin	4.557	6.385	1.401	69.671	84.822	1.217	2.618	Indifferent
Tylo + Diclofenac	4.557	7.656	1.679	70.871	72.917	1.029	2.708	Indifferent
Tylo + Ivermectin	5.857	6.385	1.090	41.016	59.926	1.461	2.551	Indifferent
Oxy + Meloxi	31.25	6.844	0.219	125	31.25	0.25	0.469	Synergistic*
Oxy + Ketoprofen	31.25	12.213	0.391	69.671	35.435	0.509	0.900	Additive
Oxy + Flunixin	5.857	8.859	1.513	69.671	71.769	1.030	2.543	Indifferent
Oxy + Diclofenac	6.844	14.648	2.140	125	62.5	0.5	2.640	Indifferent
Oxy + Ivermectin	6.844	17.932	2.620	77.247	125	1.618	4.238	Antagonistic
Chlor + Meloxi	6.569	18.229	2.775	40.994	77.247	1.884	4.659	Antagonistic
Chlor + Ketoprofen	14.648	10.926	0.746	77.247	84.822	1.098	1.844	Indifferent
Chlor + Flunixin	12.109	6.569	0.543	125	55.768	0.446	0.989	Additive
Chlor + Diclofenac	15.625	13.055	0.836	70.871	59.927	0.846	1.682	Indifferent
Chlor + Ivermectin	15.625	12.109	0.775	37.464	54.662	1.459	2.234	Indifferent
Vanco + Meloxi	14.648	25.982	1.774	77.247	71.769	0.929	2.703	Indifferent
Vanco + Ketoprofen	12.109	14.648	1.209	70.871	35.435	0.5	1.709	Indifferent
Vanco + Flunixin	8.859	6.844	0.773	69.671	125	1.794	2.567	Indifferent
Vanco + Diclofenac	12.109	14.648	1.209	125	77.247	0.618	1.827	Indifferent
Vanco + Ivermectin	7.813	11.248	1.439	54.663	72.917	1.334	2.773	Indifferent
Cipro + Meloxi	8.859	11.248	1.269	59.926	84.822	1.415	2.684	Indifferent
Cipro + Ketoprofen	5.857	6.844	1.169	41.016	62.5	1.524	2.693	Indifferent
Cipro + Flunixin	6.569	7.656	1.165	77.247	59.926	0.776	1.941	Indifferent
Cipro + Diclofenac	8.859	3.906	0.441	117.188	54.662	0.466	0.907	Additive
Cipro + Ivermectin	12.109	18.229	1.505	41.016	117.187	2.857	4.362	Antagonistic

FIC <0.5 Synergistic*, 0.5–1.0 Additive, 1.0–4.0 Indifferent, >4.0 Antagonist.

resistive order of isolates was as; “cefoxitin > gentamicin > amoxicillin > oxacillin > trimethoprim + sulfameth oxazole > vancomycin > oxytetracycline > tylosin > fusidic acid > chloramphenicol > ciprofloxacin ≥ levofloxacin ≥ amikacin ≥ moxifloxacin ≥ linezolid.

3.4. Well diffusion assay

Zones of inhibition (ZOI) of antibiotics showed significant variation ($p = 0.017$) when compared within the group. Highest ZOI was observed for chloramphenicol (15.53 ± 3.722), ciprofloxacin (15.00 ± 2.80), and tylosin (15.67 ± 1.527) while sulfamethoxazole (11.00 ± 1.04) and vancomycin (11.00 ± 1.480) exhibited the least ZOI (Table 3). Non-antibiotics ZOI comparison, when tested against selected isolates, showed a significant response ($p = 0.001$) within the group (Table 3). Flunixin meglumine (13.33 ± 1.527) and ketoprofen (10.40 ± 0.964) showed the highest ZOI while ivermectin (5.67 ± 1.527) showed the lowest ZOI.

Antibiotics (oxytetracycline, sulfamethoxazole, and chloramphenicol) showed non-significant variation ($p > 0.05$) when tested in combination with non-antibiotics while amoxicillin, gentamicin, tylosin, vancomycin, and ciprofloxacin elicited the significant response ($p < 0.05$) in combination with non-antibiotics against tested isolates (Table 4).

An increase (%) in ZOI of antibiotics when used in combination with non-antibiotics was observed highest for vancomycin, ciprofloxacin, and gentamicin (Fig. 2). The highest ZOI increase was noted for vancomycin in combination with flunixin (145.45%) and meloxicam (139.36%);

gentamicin with flunixin (85.71%) and ciprofloxacin with ivermectin (71.13%).

3.5. Synergy evaluation of antibiotics and non-antibiotics

Fractional inhibitory indices of antibiotics and non-antibiotics revealed synergistic behavior in the combination of gentamicin with ketoprofen; sulfamethoxazole and oxytetracycline with meloxicam whereas antagonism was observed for the combination of chloramphenicol with meloxicam; oxytetracycline and ciprofloxacin with ivermectin. An additive effect was noted for the combination of amoxicillin and gentamicin with meloxicam; amoxicillin and chloramphenicol with flunixin; and sulfamethoxazole and ciprofloxacin with diclofenac sodium while remaining combinations showed an indifferent response in cadre (Table 5).

3.6. Hematobiochemical profile of MRSA-infected rabbit model

Pre- and post-inoculation hematological analysis showed non-significant differences ($p > 0.05$) in white blood cells (WBCs) among infected and non-infected groups (Fig. 3). However, a significant difference ($p < 0.05$) in WBCs counts was found among all the groups on day 3 post-inoculation. While a non-significant difference ($p > 0.05$) in RBCs and hemoglobin parameters was noted among infected and non-infected groups pre- and post-inoculation. However, lymphocyte count showed a significant difference ($p < 0.05$) between groups pre- and post-inoculation of MRSA. In the serum biochemistry profile, bilirubin

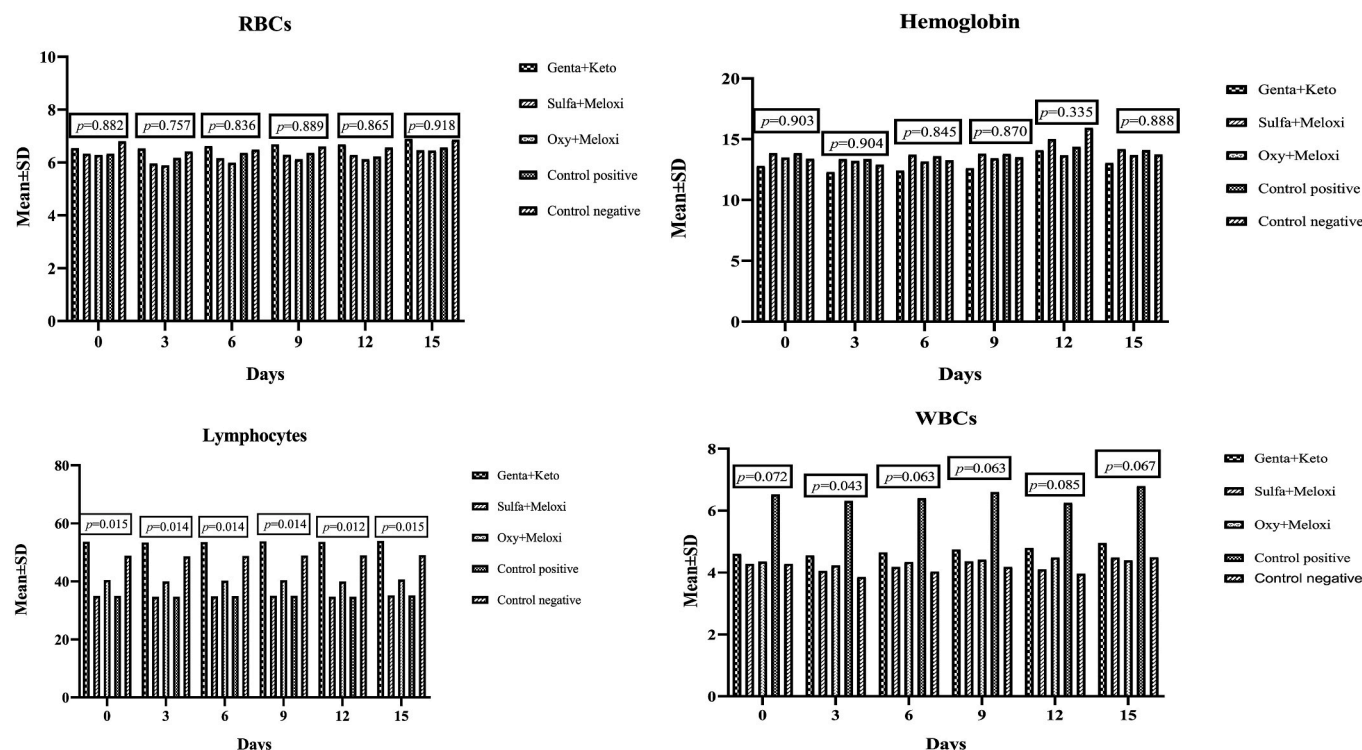


Fig. 3. Hematological parameters of MRSA infected rabbits (a) Pre- and post-inoculation Red Blood Cell concentrations for groups 1, 2, 3, 4, and 5. RBCs were slightly high in groups 1 and 5 as compared to other groups with a non-significant difference between groups. (b) Pre- and post-inoculation hemoglobin levels in experimental animals showed increased hemoglobin in control negative group on day 12. (c) Pre- and post-inoculation lymphocyte count in experimental animals showing an increased lymphocyte count in MRSA infected group 1 as compared to other groups (d) Pre- and post-inoculation WBCs count in experimental animals showing an increased WBCs counts in MRSA infected groups (1, 2, 3 and 4) as compared to group 5. Values are expressed as Mean $\pm$ Std and $p < 0.05$ indicates a significant difference.

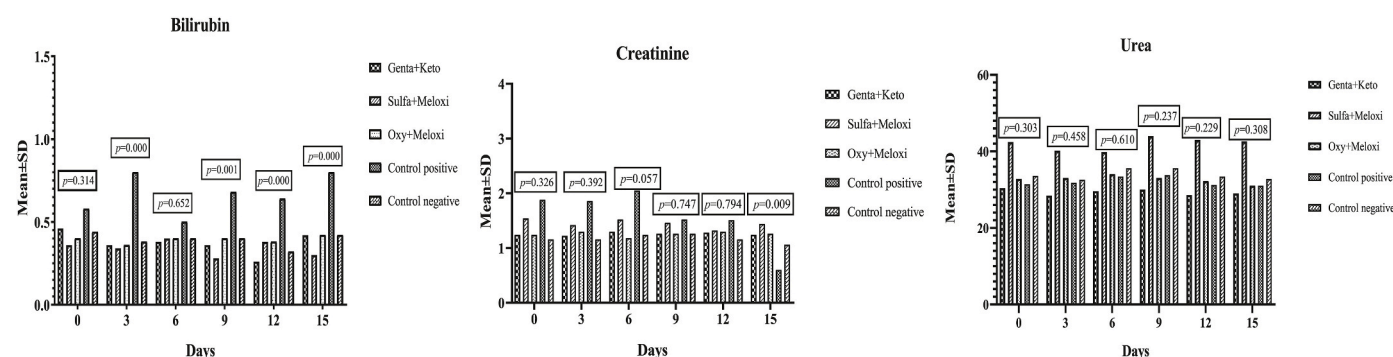


Fig. 4. Serum biochemistry profile of MRSA challenged rabbits (a) Pre- and post-inoculation bilirubin concentrations for groups 1, 2, 3, 4, and 5. Bilirubin was very high in control positive group as compared to other groups with a significant difference between groups on day 3, 9, 12, and 15. (b) Pre- and post-inoculation creatinine levels in experimental animals showed increased creatinine in control positive group except on day 15. (c) Pre- and post-inoculation urea count in rabbits showed an increased urea count in MRSA-infected group 2 as compared to other groups with non-significant differences between groups. Values are expressed as Mean $\pm$ Std and $p < 0.05$ indicates a significant difference.

showed a significant difference between groups on days 3, 9, 12, and 15 except non-significant difference ($p > 0.05$) on days 0 and 6 (Fig. 4).

3.7. Histopathological changes in skin tissue

Characteristic gross lesions observed in the skin were the presence of edema, erythema, nodules, skin elevation, ulceration, and dermonecrosis. Hyperplasia of the epithelium was observed along with the presence of mitosis in the basal layer. In the H&E-stained histological section of rabbit skin, hyperplasia, vascular dilatation, sloughed-off epidermis (at some places), hyperkeratosis, infiltration of inflammatory cells, and hemorrhages were observed. While histopathological findings in the

dermis of infected skin were the proliferation of fibrous connective tissue, edema in the superficial dermis, and vascular dilatation (Fig. 5).

4. Discussion

The current study demonstrated the presence of antimicrobial-resistant strains of *S. aureus*, especially MRSA in bovine mastitis cases highlighting the opportunities for further dissemination and transmission of such bacteria in the coming future. The current study reported a high occurrence rate of *S. aureus* from bovine mastitis. In Pakistan, previous studies have reported the presence of *S. aureus* in bovine mastitis with prevalence ranging between 10 and 65% and even

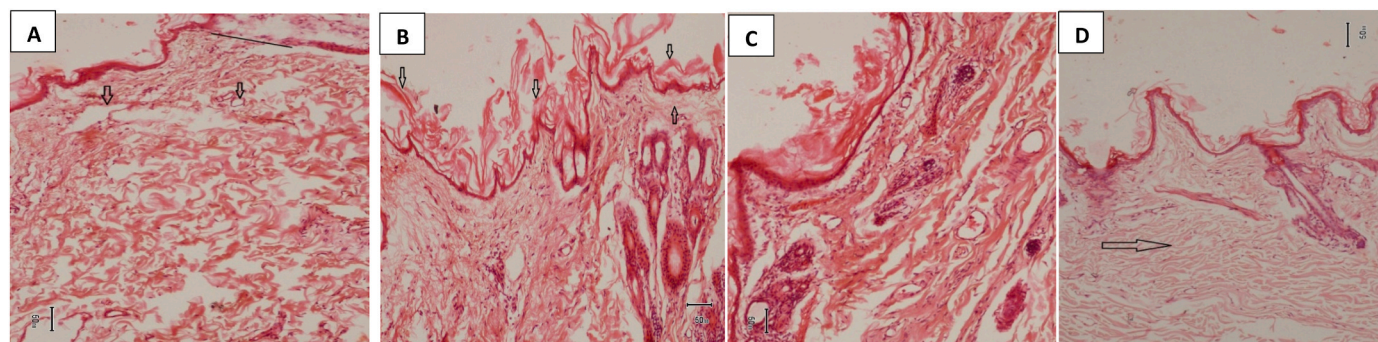


Fig. 5. Histopathology of MRSA infected skin of rabbits (A) Sloughed off epidermis and vascular dilatation, (B) Hyperkeratosis can be seen, (C) Infiltration of inflammatory cells and hemorrhages, (D) Proliferation of fibrous connective tissue in the superficial dermis.

up to 85% [29,30]. Different sample sizes, management variations between urban and rural locations, and geographic variables all account for significant variation in prevalence rates in the region [31–33].

Although the expression of *mecA* gene is considered an important mechanism responsible for methicillin resistance in *S. aureus*, some other mechanisms such as hyperproduction of beta-lactamase and changes in amino acids profile of Penicillin-binding protein cascade (PBPs 1, 2, and 3) by *S. aureus* could be the basis of beta-lactam resistance [34–36]. The *mecA* gene of *S. aureus* is considered a housekeeping gene [37]. The current study depicted a low prevalence of the *mecA* gene in *S. aureus* isolates but it provides an opportunity to search for other intrinsic factors which can compete with the *mecA* gene in generating antimicrobial resistance in MRSA endemic regions.

The current study sequences revealed high similarity with *mecA* gene of MRSA from NCBI database. Agricultural intensification, habitat invasion, and domestication of animals are potential factors that make it easier for bacterial diseases to transfer from one host to another [38]. *S. aureus* is typically spread through mechanical vectors or direct skin-to-skin contact [39]. Moving animals with MRSA colonies from one farm to another, direct contact with colonized humans, and animal contact with contaminated transport vehicles are the most significant ways to spread MRSA in livestock systems [40].

Current study isolates showed high resistance towards cefoxitin, amoxicillin, and oxytetracycline while higher susceptibility towards linezolid, gentamicin, and moxifloxacin. These study findings are in line with [41] and contrary to studies reported by Ref. [42]. The sensitivity pattern of *S. aureus* isolates to cefoxitin while resistance to oxacillin in the current study is in line with findings reported by Ref. [30] showing variation in antibiotic sensitivity assay against both antibiotics.

Synergy testing of antibiotics in combination with non-antibiotics showed synergistic effect for some combinations of antibiotics with NSAIDs. The synergistic effect of antibiotics with NSAIDs against MRSA are well studied previously by Refs. [22,24,43,44] which may be due to targeting various sites of bacteria [24]. NSAIDs exhibit anti-bacterial action against *S. aureus* by blocking *agrA*-regulated virulence, preventing hemolysis, inhibiting staphyloxanthin production, and down-regulating the expression of *fnbA* and *icaA* genes, which are critical in biofilm formation [45–47]. NSAIDs are the drugs frequently prescribed along with antibiotics to decrease pain and fever due to bacterial infections [48]. Diclofenac sodium and meloxicam cause negative regulation of the *icaA* gene expression, which encodes a major constituent of the polysaccharides from the extracellular matrix, the polysaccharide intercellular adhesion, which is the key component of the EPS matrix of *S. aureus* [49].

In addition to NSAIDs, ivermectin has also been reported as having bacteriostatic properties specifically against *S. aureus* [50]. Other studies have also reported variations in *in-vitro* and *in-vivo* results [51–53]. Temperature fluctuations, the influence of the cell-matrix on the drug effect, cell binding pathways [54], optimization and validation

of the *in-vivo* model for minimization of individual variations, local immunogenic response, enzymatic degradation, stabilization of activity [52], pharmacodynamics and pharmacokinetic characteristics of living body [53] are the necessary factors causing changes in laboratory and *in-vivo* results; so it becomes hard to attain reproducibility and repeatability of the results [55].

The current study has reported an increased amount of white blood cells with significant differences between groups as an indicator of infection. White blood cells are released to fight against infection. In most bacterial infections, there is leukocytosis along with neutrophilia [56]. The increased amount of lymphocyte count post inoculation indicates immunological stress in animals in response to infection [57]. In rabbits, the primary granulocytic leukocyte is commonly known as heterophil instead of neutrophil [58]. Following lymphocytes, heterophils are considered the second most frequently present leukocyte in the rabbit [59]. The presence of cells in the lesion of rabbits suffering from *Staphylococcal* mastitis can give an idea about mastitis chronicity, with larger numbers of macrophages, T lymphocytes, and plasma cells in undeveloped lesions [60].

Histopathological analysis is an important tool for evaluating tissue damage as a result of bacterial invasion. In the current study, hyperplasia of epithelium, vascular dilatation, sloughed off epidermis, hyperkeratosis, infiltration of inflammatory cells, and hemorrhages were observed. Similar histopathological findings in *S. aureus* infected rabbit skin model have been reported by Ref. [61]. Excessive polymorphonuclear cells can damage the epithelium by respiratory burst and degranulation causing the release of granular enzymes [62]. Previous studies in rabbit models of dermonecrosis showed that some toxins, such as alpha-toxin and phenol-soluble modulin alpha peptides, are an important determinant of virulence in the pathogenesis of *S. aureus* infection of the skin and soft tissues [63,64]. The current study provided details on resistance modulation strategies against MRSA infections and suggest conducting field trials for further validation of these therapeutics in treating bovine mastitis.

5. Conclusion

The current study was designed to characterize methicillin-resistant *S. aureus*, its pattern of resistance against antibiotics as well as the use of resistance modulation strategies for providing potential therapeutic plans to fight against mastitis. Current study findings suggest that some other mechanisms are also responsible for beta-lactam resistance of MRSA except for the presence of *mecA* gene, so, a point must be considered that only molecular confirmation by PCR is not sufficient for confirmed characterization of MRSA isolates. *In-vitro* successful combination of drugs (Gentamicin with ketoprofen; sulfamethoxazole with meloxicam and oxytetracycline with meloxicam) were tested by inoculating MRSA skin infection in rabbits. On histopathological examination, hyperkeratosis of epithelium along with inflammatory changes was

observed in MRSA-infected rabbits. The current study will help in the reversal of antibiotic resistance by using a combination of antibiotics with non-antibiotics as well as to control mastitis through an antimicrobial stewardship program.

CRedit authorship contribution statement

Iqra Muzammil: Writing – original draft, Data curation. **Muhammad Ijaz:** Conceptualization. **Muhammad Hassan Saleem:** Writing – review & editing. **Muhammad Muddassir Ali:** Methodology, Formal analysis.

Declaration of competing interest

All authors have declared no conflict of interest in the submission/publication of this data.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.micpath.2022.105691>.

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